**FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 4**

**PHYSICS / APPLIED PHYSICS**

**OPTICS**

Module 3 & Module 4 (Unit 17): Diffraction and Introduction to Polarisation

| **Maximum Time: 2 Hours** | | **Maximum Marks: 70** |
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| | **Section A** | |
| | **All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)** | |
| 1 | Explain physically, by dividing the slit into two equal halves, why the first minimum in a single-slit Fraunhofer diffraction pattern occurs at $b\sin\theta = \lambda$. | **3** |
| 2 | Give the expression for the intensity distribution in a two-slit Fraunhofer diffraction pattern and explain the physical significance of the two factors involved. What happens when the slit width $b$ becomes very small? | **3** |
| 3 | What do you mean by missing orders in a double-slit Fraunhofer diffraction pattern? State the condition under which a particular interference maximum is absent. | **3** |
| 4 | Give the grating equation and explain why no dispersion occurs in the zeroth-order spectrum. What is meant by overlapping of spectra in a diffraction grating? | **3** |
| 5 | What do you mean by the resolving power of a grating? Give an expression for the same explaining the terms involved. | **3** |
| 6 | How does Fresnel diffraction differ from Fraunhofer diffraction? For a plane wavefront of monochromatic light, what is the difference in areas of the $n^{\text{th}}$ and $(n-1)^{\text{th}}$ Fresnel half-period zones? | **3** |
| 7 | Write down the equation for the principal focal length of a zone plate. Compare a zone plate and a convex lens. | **3** |
| 8 | Define the Fresnel integrals $C(\tau)$ and $S(\tau)$ and state their basic properties. What is Cornu's spiral? | **3** |
| 9 | Define polarization of light. Distinguish between unpolarized, linearly polarized, and circularly polarized waves. | **3** |
| 10 | Write down the expressions for the electric and magnetic field vectors $\mathbf{E}$ and $\mathbf{H}$ of a linearly polarized plane wave propagating along the $z$-direction in a dielectric. Define polarizer and analyser. | **3** |
| | **Section B** | |
| | **All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)** | |
| 11 | A parallel beam of light of wavelength $\lambda = 5 \times 10^{-5}\ \text{cm}$ is incident normally on a narrow slit of width $0.2\ \text{mm}$. The Fraunhofer diffraction pattern is observed on a screen placed at the focal plane of a convex lens of focal length $20\ \text{cm}$ placed close to the slit. Calculate the distance between the first two minima and the first two secondary maxima on the screen. | **6** |
| 12 | Derive the expression for the resultant electric field and intensity distribution in the Fraunhofer diffraction pattern produced by two parallel slits, each of width $b$ and separated by a distance $d$. Show that when $b = d$, the resulting diffraction pattern corresponds to that of a single slit of width $2b$. | **6** |
| 13 | A diffraction grating having $15,000$ lines per inch is illuminated normally by sodium light ( $\lambda_1 = 5890\ \text{AA}$ and $\lambda_2 = 5896\ \text{AA}$). Calculate: (i) the angular separation of the $D_1$ and $D_2$ lines in the second-order spectrum, and (ii) the linear separation between the two lines in the first-order spectrum observed on the focal plane of a convex lens of focal length $10\ \text{cm}$. | **6** |
| 14 | With a neat figure, explain the construction of Fresnel's half-period zones on a plane wavefront. Show that the areas of all half-period zones are approximately equal to $\pi\lambda d$, and prove that the resultant amplitude at an axial point due to the entire wavefront is half the amplitude produced by the first half-period zone. | **6** |
| 15 | Explain the principle and construction of a Fresnel zone plate and show that it produces multiple foci. Consider a zone plate with radii $r_n = 0.1\sqrt{n}\ \text{cm}$; for $\lambda = 5 \times 10^{-5}\ \text{cm}$, calculate the positions of the principal and secondary foci. | **6** |
| 16 | Obtain the expression for the intensity distribution on a screen due to Fresnel diffraction of a plane wave incident normally on a straight edge in terms of the Fresnel integrals $C(v_0)$ and $S(v_0)$. Hence show that the intensity on the edge of the geometrical shadow is $\frac{1}{4}I_0$. | **6** |

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| 17 | Draw the intensity distribution curve of Fresnel's diffraction at a straight edge and explain its main features. Consider a straight edge illuminated by a parallel beam of light with $\lambda = 6 \times 10^{-5}$ cm; calculate the positions of the first two maxima and the first minimum on a screen at a distance of 50 cm from the edge (given $v_0 = -1.22, -2.34$ for the first two maxima and $v_0 = -1.87$ for the first minimum). | **6** |
| 18 | With suitable diagrams, explain how the experiment with transverse waves on a string passing through two narrow slits and the corresponding optical experiment with two Polaroids demonstrate the transverse wave nature of light. Why cannot longitudinal waves exhibit polarization? | **6** |
| **Section C** | | |
| **Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)** | | |
| 19 | Discuss the Fraunhofer type of diffraction produced by a narrow single slit illuminated normally by monochromatic light. Derive the expression for the intensity distribution, obtain the conditions for maxima and minima, explain the graphical method for finding the positions of secondary maxima, and draw a diagram indicating the intensity distribution in the diffraction pattern. | **10** |
| 20 | Derive the expression for the intensity distribution in Fraunhofer diffraction due to $N$ identical parallel slits. Discuss the positions of principal maxima, secondary minima, and absent spectra, and derive the equation for the angular half-width of the principal maxima to show that they become sharper as $N$ increases. | **10** |